Case 12: Expertise and automation

1 Context

The introduction of computers to the workplace revolutionized how clerical tasks were performed by white collar professionals. Routine tasks that could be effectively codified became increasingly performed by computers as workers and firms adapted to the possibilities of new technologies. And so, computers took over many of the calculations and record-processing activities once performed by accountants, as well as by many activities performed by inventory clerks.

Computers made it so employees could complete their work with less time and effort. This led to changes in the task-content of white collar jobs. For some, their works started to place more importance in tasks that required a higher degree of *worker expertise*; for others the remaining tasks moved away from *expertise* and into manual work. Accountants continued to investigate discrepancies, interpret financial information, and advise clients. Inventory clerks increasingly concentrated on counting, moving, and checking physical stocks.

This distinction matters for wages. Expertise is valuable partly because it allows a worker to perform activities that others cannot perform. Technological innovation can make occupations more productive and in doing so it might make the expertise of workers more valuable, or it can replace it. Doing away with the role for *worker expertise* removes barriers to the entry of new workers to the occupation; effectively, anyone can now make it into the occupation. This lowers the wages of incumbent workers, but can be beneficial to the workers who now have access to occupations previously inaccessible to them.

This case study is divided in two parts. Part I develops a task-assignment model of expertise where workers choose occupations as automation changes the role of worker expertise in them. This model helps us frame and understand Part II, that examines how changes in the task content of U.S. occupations relate to wages and employment between 1980 and 2018. The seven questions carry 120 marks: 80 for the model and 40 for the evidence.

Part I. Solving the model

2 Workers, tasks, and occupations

There is a total workforce $\overline{L}$. Each worker supplies one unit of time and has expertise $e \in (0, 1)$. Expertise is uniformly distributed. Workers choose the highest paying occupation they can perform.

Tasks are indexed by $t \in (-\theta, 1)$, where $\theta > 1$. The value of t represents the required experience to perform a task, so a worker can perform task t if $t \leq e$. Conditional on being able to perform a task, every worker has the same productivity: one unit of time produces one unit of that task's services. All workers can perform tasks $t \in (-\theta, 0)$ for which no experience is required. We call those tasks *generic*.

An occupation is indexed $\varphi \in (0, 1)$, which determines the range of tasks that comprises it

$$\mathcal{T}_\varphi = [-\theta(1-\varphi), \varphi], \qquad T_\varphi = \theta(1-\varphi) + \varphi. \tag{1}$$

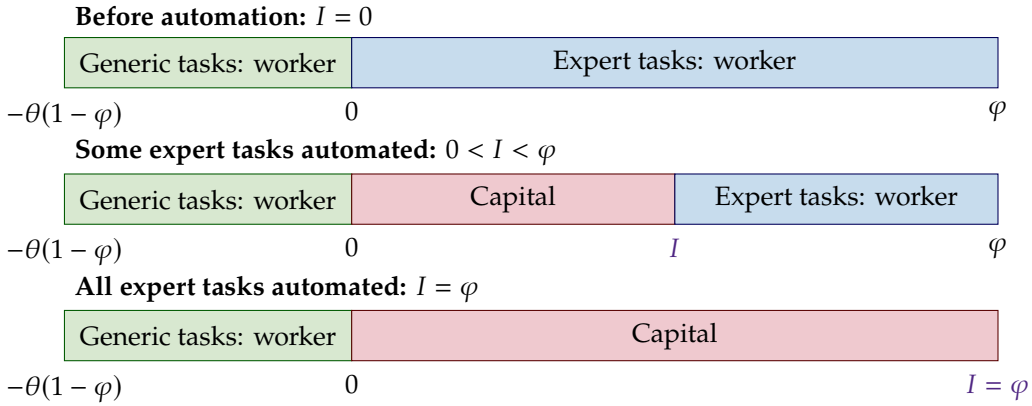
Here T_φ is the length of the occupation's task interval. Occupations with higher φ require more demanding expert tasks and contain fewer generic tasks. The bundle is fixed for each occupation. In order to work in the occupation a worker needs experience $e \geq \varphi$ to be able to perform all of its tasks..

3 Automation, tasks, and expertise

Automation technology can change expertise requirements. Let the *automation frontier* be $I \in (0, 1)$. Capital can perform tasks $0 < t \leq I$, producing $\eta > 1$ units of task services per unit of capital. It cannot perform generic tasks or tasks above I, so I is akin to the level of expertise of the machines.

Assume all tasks within its reach are assigned to capital. This is profitable in the equilibrium considered below, where the cost of a unit of machine task services, r/η, is below the lowest wage. So, any occupation with $\varphi < I$ has no expert tasks left for workers (all have been automated) and any worker can work in them, while occupations with $I < \varphi$ need workers to perform expert tasks $t \in (I, \varphi)$.

Exhibit 1. Follow one occupation as automation advances

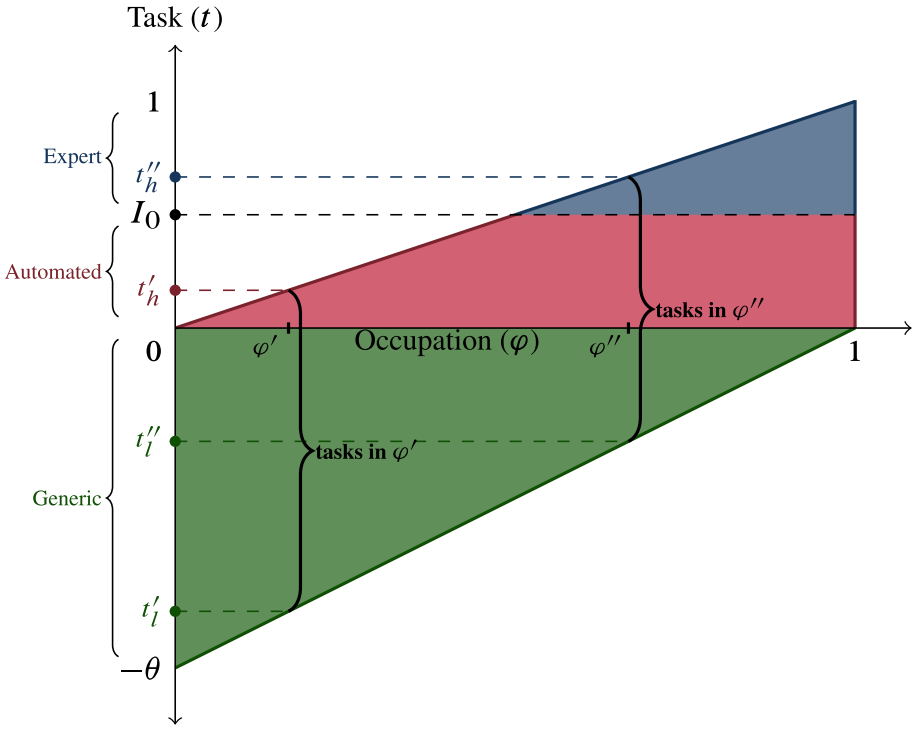


Illustrative model diagram. All three rows contain the same occupation's tasks. Only the automation frontier changes. Green denotes generic human tasks, blue expert human tasks, and red automated tasks.

Exhibit 1 holds the occupation's task bundle fixed while the automation frontier advances. In the first two rows, a worker still needs enough expertise to perform the occupation's highest task, φ. Automating some easier tasks does not remove that requirement. In the third row, capital performs every expert task and the worker supplies the generic activities. The job still needs a person, but no longer requires expertise above the generic level.

Exhibit 2 holds the frontier at I_0 and compares occupations with different task bundles. In occupation φ', capital performs every task that requires expertise. Workers still perform its generic activities. Occupation φ'' also uses capital, but retains expert tasks above I_0. Moving the frontier changes the division of work within each fixed bundle. It can also change who is qualified to enter an occupation.

Exhibit 2. Compare occupations at a given frontier



Model diagram. The horizontal axis orders occupations; the vertical axis orders tasks. A vertical slice contains one occupation's bundle. Green tasks remain human, red tasks are automated, and blue tasks require human expertise. I_0 denotes the current automation frontier. The labels t_l and t_h mark the lower and upper endpoints of the two example bundles.

4 Task assignment and labor demand

The output in occupation φ is a Cobb–Douglas aggregate of task output:

$$Y_\varphi = T_\varphi \exp\left\{\frac{1}{T_\varphi}\int_{\mathcal{T}_\varphi} \log y(t)\,\mathrm{d}t\right\}, \tag{2}$$

where task output is $y(t) = \ell(t)$ on human tasks and $y(t) = \eta\kappa(t)$ on automated tasks.

A worker in occupation φ supplies one unit of labour and employs k units of capital. Let $\ell(t)$ be time allocated to a human task and $\kappa(t)$ capital allocated to an automated task. The resource constraints are

$$\int_{\mathcal{H}_\varphi} \ell(t)\,\mathrm{d}t = 1, \qquad \int_{\mathcal{M}_\varphi} \kappa(t)\,\mathrm{d}t = k, \tag{3}$$

where $\mathcal{H}_\varphi$ and $\mathcal{M}_\varphi$ are the human and automated parts of the bundle. Write m_φ for the measure of automated tasks and $h_\varphi = T_\varphi - m_\varphi$ for the measure of human tasks.

It is optimal to divide inputs evenly across the tasks they perform. This happens because all tasks have the same weight in the Cobb-Douglas aggregate, so the resource constraints have one common Lagrange

multiplier for all tasks using that resource.

Question 1. Who can produce, and how much? (20 marks)

(a) Find m_φ and h_φ. State which workers can enter an occupation when $\varphi > I$ and when $\varphi \leq I$. [6 marks]

(b) Solve the worker's allocation problem at a given $k > 0$. Find the time and capital used per task and substitute them into equation (2). [8 marks]

(c) Let $\alpha_\varphi = m_\varphi / T_\varphi$. If an occupation employs L_φ qualified workers and K_φ units of capital, show that its output can be written as

$$Y_\varphi = \left(\frac{L_\varphi}{1-\alpha_\varphi}\right)^{1-\alpha_\varphi} \left(\frac{\eta K_\varphi}{\alpha_\varphi}\right)^{\alpha_\varphi}.$$

Explain why capital is divided equally among these workers. [6 marks]

Question 2. Comparative statics as automation advances (20 marks)

Fix an occupation φ and its task bundle T_φ. In parts (a) and (b), hold capital per worker at $k > 0$ and consider $0 < I < \varphi$.

(a) As I rises, find the changes in m_φ, h_φ, and $\alpha_\varphi = m_\varphi / T_\varphi$ per unit increase in I. How do the worker's time per human task and capital per automated task change? [8 marks]

(b) Let $q_\varphi(k, I)$ denote the output of one qualified worker from Question 1(b). Show that

$$\frac{\partial \log q_\varphi(k, I)}{\partial I} = \frac{1}{T_\varphi} \log\left(\frac{\eta k h_\varphi}{m_\varphi}\right).$$

Under what condition does output per worker rise at fixed k? Explain why a larger automated task range alone does not determine the sign. [7 marks]

(c) What happens when I reaches and then exceeds φ? Distinguish the continuity of task measures and fixed-k output from the change in which workers qualify for the occupation. [5 marks]

5 Prices and the demand for workers

Use the occupational production function in Question 1(c) for the remaining model questions. Firms hire qualified workers at wage w_φ and rent capital at price r. Competition and constant returns imply that the output price p_φ equals unit cost.

The economy combines occupational outputs into a final good:

$$Y = \left[\int_0^1 Y_\varphi^{(\lambda-1)/\lambda} \, \mathrm{d}\varphi\right]^{\lambda/(\lambda-1)}, \qquad \lambda > 1. \tag{4}$$

Here λ is the elasticity of substitution between occupational outputs. It is distinct from substitution between tasks within a worker's job. Normalize the final-good price to one. Occupational output

demand and the corresponding price index satisfy

$$Y_\varphi = Y p_\varphi^{-\lambda}, \qquad \int_0^1 p_\varphi^{1-\lambda}\,\mathrm{d}\varphi = 1. \tag{5}$$

The wage and rental rate are measured in units of the final good.

Question 3. From production to labour demand (20 marks)

(a) Derive the cost-minimizing capital–labour ratio and the unit cost of occupational output. Show that

$$p_\varphi = w_\varphi^{1-\alpha_\varphi}(r/\eta)^{\alpha_\varphi}.$$

What shares of occupational revenue go to labour and capital? [10 marks]

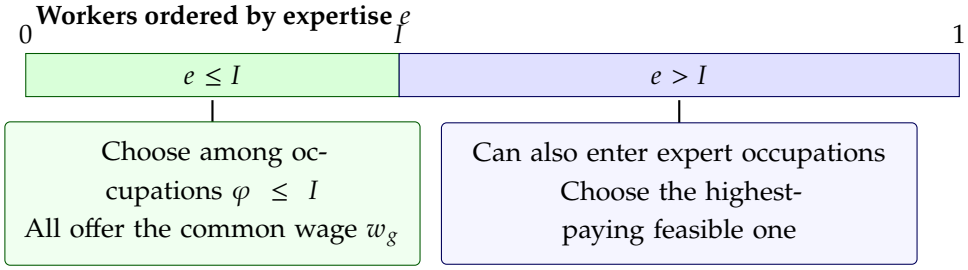
(b) Combine the firm's labour demand with equation (5) to obtain

$$L_\varphi = (1-\alpha_\varphi) Y w_\varphi^{-d_\varphi}(r/\eta)^{\alpha_\varphi(1-\lambda)}, \qquad d_\varphi = \lambda - \alpha_\varphi(\lambda - 1).$$

Holding α_φ, Y, and r/η fixed, explain why labour demand slopes down with the wage. [10 marks]

6 Occupational choice and the automation frontier

Consider the productive-automation equilibrium in which wages are strictly increasing with φ among occupations with $\varphi > I$. This means that a worker with expertise $e > I$ will choose to work in occupation $\varphi = e$, the highest-expertise occupation available to them. All occupations with $\varphi \le I$ are accessible to every worker and are chosen by workers with expertise $e \le I$. Free movement between these occupations gives them a common wage w_g. Assume capital productivity is high enough that $r/\eta < w_g$ throughout the comparisons.



Illustrative model diagram. The expertise distribution is fixed. Raising I moves the boundary between the two groups to the right. Higher-expertise workers retain the option of a generic occupation, but prefer an expert occupation when it pays more.

Once occupational choice is determined, market clearing pins down Y, r, and w_g. The economy must employ its workforce $\overline{L}$, use its capital stock $\overline{K}$, and satisfy the price-index condition in equation (5). For

capital, the revenue-share result gives

$$K_\varphi = \frac{\alpha_\varphi Y p_\varphi^{1-\lambda}}{r}, \qquad \int_0^1 K_\varphi \, \mathrm{d}\varphi = \overline{K}. \tag{6}$$

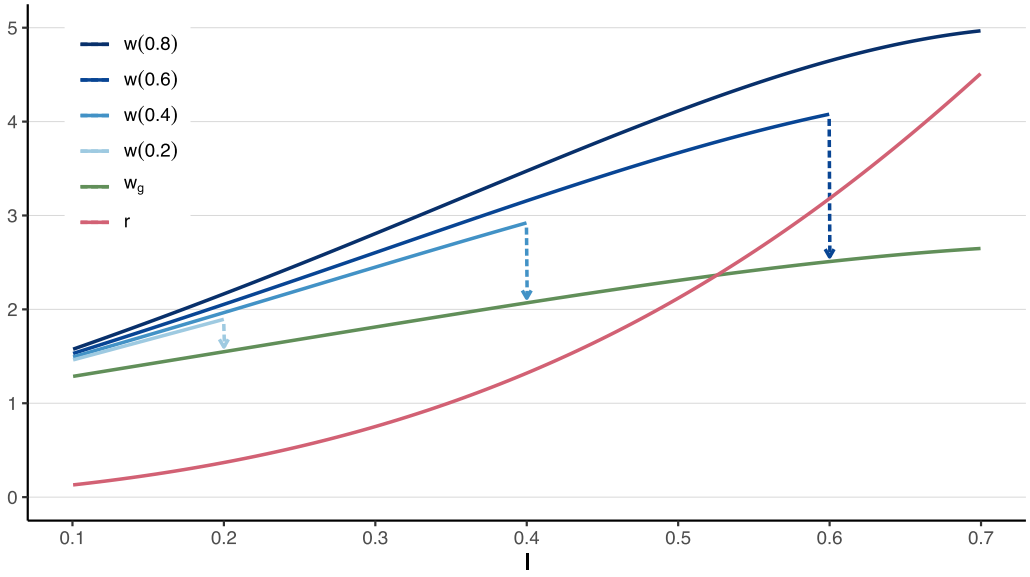
Question 4. Entry, wages, and employment (20 marks)

(a) Determine the occupation chosen by a worker with $e > I$. What is the employment density L_φ in occupations above I, and how many workers must the occupations below I employ in total? Use these results and Question 3 to write the equations that determine w_φ, w_g, r, and Y. You do not need to solve the integrals. [8 marks]

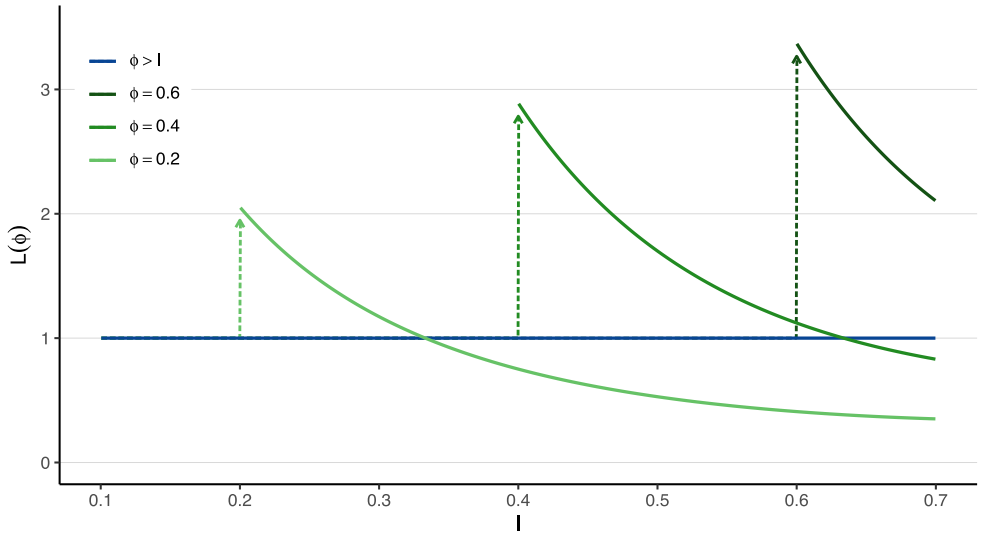
(b) Exhibit 3 shows the resulting equilibrium as I rises. Explain what happens when the frontier reaches an occupation's most demanding task, and contrast this with an increase in I that leaves that task beyond the frontier. Evaluate the claim: "If automation raises output per worker, the occupation's existing workers must receive a higher wage." [12 marks]

Exhibit 3. Advancing automation in the model

(a) Real wages and the capital rental rate



(b) Employment in selected occupations



Model simulation. The horizontal axis is the automation frontier I. In panel (a), blue curves track wages in selected occupations, green is the common wage w_g, and red is the capital rental rate r. In panel (b), the blue line shows occupations that remain expert; green curves track selected occupations after entry opens. $L(\varphi)$ is an employment density: employment in a small interval is approximately its density times the interval's width. The workforce and capital stock are each normalized to one. Other parameters are $\lambda = 4$, $\theta = 3/2$, and $\eta = 15$. No calculations using these values are required.

The panels describe allocations of a fixed workforce. They show how an occupation's employment can expand at the same point that its wage falls. After entry becomes unrestricted, its wage follows the common wage w_g. The remaining expert occupations continue to require workers who can perform their highest tasks.

Part II. Expertise in the data

7 Measuring the content of work

To examine how expertise has changed as technology has evolved, we need information about what workers do, not just how many people have each job. Auto and Thompson use the detailed descriptions of occupations from the 1977 *Dictionary of Occupational Titles* and the 2018 *O*NET* database to identify tasks within occupations and evaluate how much expertise they require. These descriptions are linked to wages and employment for 303 consistent U.S. occupations between 1980 and 2018.

An *expertise score* measures how specialized the language of a task is. The idea is that tasks that require more specialized language to be described require more expertise. The starting point is the frequency of words across school and college texts. Common words receive lower scores; words concentrated in specialized subject areas receive higher scores. For example, *luminance* is more specialized than *colour*. A familiar-word list further adjusts the scores so that ordinary words are not treated as expert language merely because they appear infrequently in textbooks. Word scores are averaged within tasks and then across the titles belonging to an occupation. This exercise provides tasks with an expertise levels that is then passed on to occupations.

However, changes in expertise for an occupation do not only come from changes in the description of tasks. They can also come from changes in the task-composition of the occupation. To evaluate this, task descriptions are then matched across dates by their meaning. A task is *retained* if a corresponding activity appears at both dates, *removed* if it appears only initially, and *added* if it appears only at the end. For a file clerk, classifying incoming records can remain part of the job even when the records are scanned rather than filed on paper. Keypunching can disappear, while entering information into a document-management system can appear as a new activity.

So, the change in expertise for an occupation o is constructed in two steps. Let $\overline{X}_{77,o}$ and $\overline{X}_{18,o}$ be the average expertise of all tasks in occupation o at each date, and let $\overline{X}^{R}_{77,o}$ and $\overline{X}^{R}_{18,o}$ be the averages for retained tasks at the corresponding date. Define

$$R_o = \overline{X}^{R}_{77,o} - \overline{X}_{77,o}, \qquad A_o = \overline{X}_{18,o} - \overline{X}^{R}_{18,o}, \qquad \Delta X_o = R_o + A_o. \tag{7}$$

Here R_o is the change attributable to task removal and A_o the change attributable to task addition. Removing low-expertise tasks raises the average expertise of the tasks that remain. Removing high-expertise tasks lowers it. The comparison uses retained and all tasks *within each date*, so rewording a retained task is not itself counted as an expertise change.

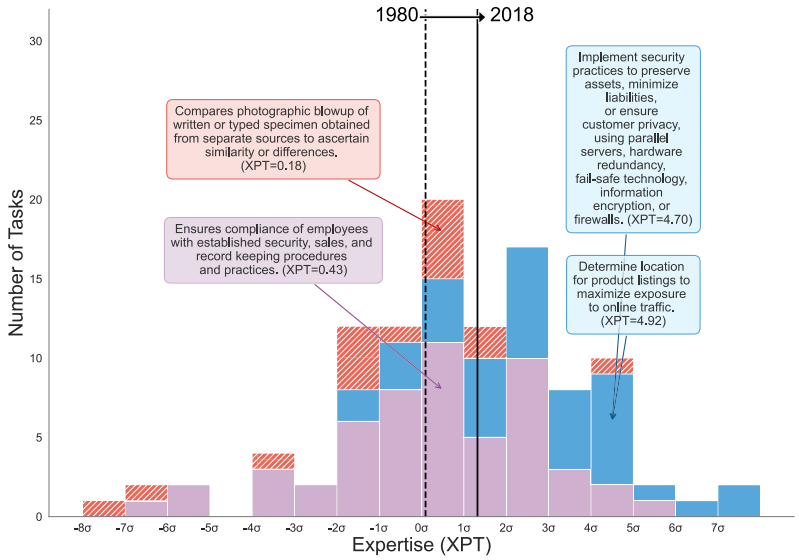
There is also a separate measure of task quantity:

$$\Delta Q_o = \frac{\text{added tasks}}{\text{all tasks in 2018}} - \frac{\text{removed tasks}}{\text{all tasks in 1977}}. \tag{8}$$

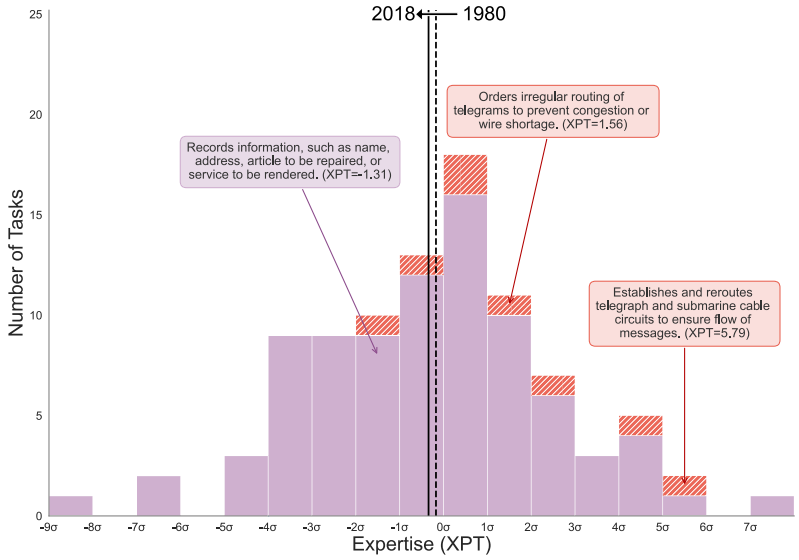
It records changes in the coverage of task descriptions, not hours spent working. An occupation can lose tasks while the expertise of its remaining work either rises or falls.

Exhibit 4. Different tasks disappear in different occupations

(a) Management support



(b) Dispatchers



Historical evidence. The horizontal axis is task expertise and the vertical axis counts task descriptions. Purple bars show retained tasks, orange hatching removed tasks, and blue bars added tasks. Vertical markers identify average expertise at the initial and final dates. Bar heights count tasks, not workers or hours.

In management support, removed tasks tend to be less expert than retained tasks, while added tasks tend to be more expert. For dispatchers, the removed tasks are comparatively expert. The loss of tasks therefore has different implications for the knowledge left in the job. The distinction is relevant to work such as assigning routes and coordinating deliveries: a system can remove demanding scheduling decisions while leaving workers to communicate instructions and monitor progress.

Question 5. What changes when tasks disappear? (15 marks)

(a) Use Exhibit 4 and equation (7) to compare the expertise change due to task removal in the two occupations. Would knowing only that both occupations lost tasks be enough to make this comparison? [8 marks]

(b) Explain why the measure in equation (7) does not simply subtract the 1977 all-task average from the 2018 all-task average. What change is it designed to leave out? [7 marks]

8 Expertise, wages, and employment

The data allows us to ask whether occupations whose work became more expert experienced different wage and employment changes. For occupation o, we can answer this question with a regression:

$$\Delta \log z_o = c + \beta \Delta X_o + \gamma \Delta Q_o + \delta_{g(o)} + u_o, \tag{9}$$

where z_o is either the occupation's wage or its employment. The regression accounts separately for task quantity: an occupation that loses demanding tasks may differ from one that gains many new tasks of the same expertise. The term $\delta_{g(o)}$ controls for common changes within broad occupational groups. All changes are expressed per decade, so an expertise change refers to standard deviations per decade, and a task-quantity change to the net share per decade.

Exhibit 5. Expertise and task quantity have different relationships with employment

	$\Delta \log$(wage)	$\Delta \log$(employment)
Expertise change: $\widehat{\beta}$	0.175	−0.510
	(0.047)	(0.199)
Task-quantity change: $\widehat{\gamma}$	0.334	1.430
	(0.123)	(0.523)
Broad occupational-group controls	Yes	Yes
Occupations	303	303

Standard errors are in parentheses. U.S. occupations, 1980–2018. Coefficients describe changes per decade.

Holding task quantity and the broad group constant, a 0.1-standard-deviation increase in expertise per decade is associated with 1.75 log points more wage growth and 5.10 log points less employment growth per decade. Holding expertise constant, a 0.10 increase in the net task-share measure is associated with 3.34 log points more wage growth and 14.30 log points more employment growth. Thus, more expert work and a larger task bundle have different employment associations, even though both have positive wage associations.

Exhibit 6. The expertise of removed and added tasks matters

Explanatory variable	Wage coefficient	Standard error
Expertise change due to removal, R_o	0.147	0.051
Expertise change due to addition, A_o	0.255	0.088
Share of initial tasks removed	−0.230	0.141
Share of final tasks added	0.549	0.206

Dependent variable: change in log wages per decade. The four variables enter together, with broad occupational-group controls and initial employment weights. There are 303 occupations. All task and expertise changes are expressed per decade.

The positive removal-expertise coefficient means that removing less expert activities has a different wage association from removing more expert activities, even at the same fraction of tasks removed. The coefficient on the removed fraction alone is negative, but is not statistically distinguishable from zero at the 5% level. A count of disappearing activities therefore does not settle how wages change.

Cheatsheet: reading the evidence
In a multiple regression, a coefficient compares outcomes holding the other included variables fixed. An increase of 0.01 in a log outcome is approximately a 1% increase in its level. A standard error measures sampling uncertainty; an approximate 95% interval is the coefficient plus or minus twice its standard error. Statistical significance does not establish causality.

Question 6. Using the model to interpret the facts (15 marks)

(a) Explain how Exhibit 5's combination of higher wage growth and lower employment growth for rising expertise can be understood through occupational entry. Relate your explanation to Exhibit 3. [8 marks]

(b) Two occupations lose the same fraction of tasks. One loses relatively undemanding tasks; the other loses relatively demanding tasks. Use Exhibit 6 to compare their predicted wage changes, holding task additions and the other included variables fixed. Does the model imply that every removal of a demanding task must cause a discontinuous wage fall? [7 marks]

Question 7. What identifies the effect of automation? (10 marks)

Does equation (9) identify the causal effect of machines on wages? Describe one competing explanation for the observed relationship and explain what additional variation or evidence would help distinguish it from the automation mechanism.

Reference

Autor, David, and Neil Thompson (2025). *Expertise*. *Journal of the European Economic Association* 23(4), 1203–1271.